

Monte Carlo Integration

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Monte Carlo Integration

Monte Carlo Integration is a statistical method based on random sampling

Idea:

- If X is a random variable with pdf $f(x)$, then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

- Thus, to estimate $\omega = \int_a^b g(x)dx$, we do

$$X_1, \dots, X_n \sim U(a, b), \text{ then } \hat{\omega} = \frac{1}{n} \sum_{i=1}^n g(X_i)$$

Algorithm

- 1) Generate iid $X_1, \dots, X_n$ from $U(a, b)$
- 2) Compute $\bar{g} = \frac{1}{n} \sum_{i=1}^n g(X_i)$
- 3) then $\hat{\omega} = (b - a)\bar{g}$

Example

Compute a Monte Carlo Estimate of $\omega = \int_0^1 e^{-x} dx$

```
set.seed(123)
n<-10000

x <- runif(n, min=0, max=1)
f_x <- exp(-x)

#Average of g(X)
omega_hat <- mean(f_x)

#compute (b-a)mean(g(X))
(1-0)*omega_hat
```

```
## [1] 0.6333057
```

Compute a Monte Carlo Estimate of $\omega = \int_2^4 e^{-x} dx$

```
set.seed(123)
n<-10000

x <- runif(n, min=2, max=4)
f_x <- exp(-x)

#Average of g(X)
omega_hat <- mean(f_x)

#compute (b-a)mean(g(X))
(4-2)*omega_hat
```

```
## [1] 0.117334
```

Estimate $\phi(-1.645) = \int_{-\infty}^{-1.645} \frac{1}{\sqrt{(2\pi)}} e^{-\frac{t^2}{2}} dt$ using Monte Carlo Method

- Generating n random samples $X_1, \dots, X_n$ from a standard normal distribution $N(0, 1)$
- Counting how many samples satisfy $X_i \leq -1.645$
- Computing the proportion:

$$\phi(-1.645) = \frac{\# \text{ of } X_i \leq -1.645}{n}$$

```
set.seed(123)
n<-10000

samples <- rnorm(n)

count <- sum(samples <= -1.645)

estimate <- count/n
estimate # or estimate <- mean(samples <= -1.645)
```

```
## [1] 0.0499
```

Importance Sampling

Suppose we want to estimate:

$$\hat{\mu} = E[g(X)] = \int g(x)f(x)dx,$$

where $X \sim f(x)$, and $f(x)$ is the density of X . We can rewrite the expectation using another density $h(x)$ (the importance distribution) as long as $h(x) > 0$ whenever $h(x)f(x) \neq 0$ (to avoid division by zero):

$$\mu = \int h(x)f(x)dx = \int g(x)\frac{f(x)}{h(x)}h(x)dx = E_h[g(x)\frac{f(x)}{h(x)}]$$

where the expectation is now taken over $X \sim h(x)$

Algorithm

The importance sampling estimator is:

1. Draw n samples $X_1, \dots, X_n \sim h(x)$
2. Compute the weights $\omega(X_i) = \frac{f(X_i)}{h(X_i)}$
3. Estimate μ as:

$$\mu = \frac{1}{n} \sum_{i=1}^n g(X_i) \frac{f(X_i)}{h(X_i)}$$

Example

Simulate $\phi(-2) = P(Z \leq -2)$ using importance sampling with $h(x) \sim \text{Cauchy}(1)$

```
set.seed(123)
n<-10000

x <- rcauchy(n)

f <- function(x){
  return(dnorm(x))
}

h <- function(x){
  return(dcauchy(x))
}

g <- function(x){
  return(x <= -2)
}

estimate <- mean(g(x) * (f(x)/h(x)))
estimate
```

```
## [1] 0.02253659
```